UNIVERSITY NAME:

Strathmore University

DEGREE COURSE:

Bachelor’s in Informatics and Computer Science

UNIT NAME AND ASSIGNMENT:

Computer simulation and modelling – Project Assignment 4.1B

LINK TO VIEW CODE:

https://github.com/misskanyi/CSM_Assignment

LECTURER'S NAME:

C. Onyango

STUDENTS NAME AND ADMISSION NUMBERS:

Kanyi Sharon Wambui - 153486

Kamau Edwin – 152803

Kariuki Rurigi Maina - 163912

Overview

This application simulates a bank with one teller serving customers one at a time. It tracks how long each customer waits, how busy the teller is, and how queues form all without needing a real bank(simulation).

Random numbers are used to generate realistic arrival and service times. The simulation then processes all customers in order and reports every timing detail.

How to Run the Application

```
# Build & run

mvn clean package

java -jar target/banking-queue-simulation-1.0.0.jar
```

1. **Configure(optional)**

Open the Configuration tab and adjust customer count, arrival, and service bounds.

If you would prefer to just generate a random sample, paste your own value or import from excel skip to part 2 below.

2. **Enter data**

Go to Simulation tab.

Click '**Generate Random'** to generate random values, you can also paste your own 100 values or import an excel file.

3. **Run**

Click "Run Simulation". Processing is instant.

4. **View results**

Results tab = full customer table.

Analytics tab = summary statistics.

5. **Export and Clear**

Click "Export to Excel" to save results as a spreadsheet.

To clear simulation or if you want to run another simulation click clear.

Simulation Parameters

The simulation runs with these fixed values:

- Customers -100
  - Per run (default)
- Inter-Arrival Time
  - 1 – 8 min
- Uniform distribution
- Service Time
  - 1 – 6 min
- Uniform distribution
- Servers
  - 1
- Single teller

How the Mathematics Works

Random numbers (0–1) are converted to real times using this formula:

Time = min + (max − min) × R

e.g. IAT: 1 + (8 − 1) × 0.43 = 4.01 minutes

For each customer the simulation then calculates

FIELD	HOW ITS CALCULATED
Arrival Time	Previous arrival + inter-arrival time
Service Start	Later of arrival time OR when teller finishes previous customer
Waiting Time	Service Start − Arrival Time
Departure Time	Service Start + Service Time
Time in System	Departure − Arrival (total visit duration)

Output Statistics Explained

STATISTICS	WHAT IT MEANS
Average Waiting Time	How long a typical customer queues before being served
Average Service Time	How long the teller spends with a typical customer
Average Inter-Arrival Time	Average gap between consecutive customer arrivals
Average Time in System	Total time in the bank wait + service combined
Server Utilization	% of time the teller was serving (not idle)
Probability of Waiting	Fraction of customers who had to wait at all
Customers Who Waited	Count of customers who arrived while teller was busy
Total Simulation Time	Minute-mark when the last customer left
Total Idle Time	Total minutes the teller had no one to serve
Average Queue Length	Average number of people waiting at any point in time
Max Queue Length	The longest the queue ever got during the run

Software Architecture

Built with Java 17 + Swing, organized into three packages. Simulation logic is fully separate from the display code.

PACKAGE	CLASS	ROLE
model	Customer	Stores all timing data for one customer
model	QueueStatistics	Holds the computed summary metrics
simulation	QueueSimulator	Core engine runs the queue, computes stats
excel	ExcelHandler	Imports random numbers from Excel; exports results
gui	MainFrame	Application window with sidebar navigation
gui	InputPanel	Data entry for random numbers
gui	TableOutputPanel	Per-customer results table
gui	StatisticsOutputPanel	Analytics dashboard with stat cards

Pages in our simulation

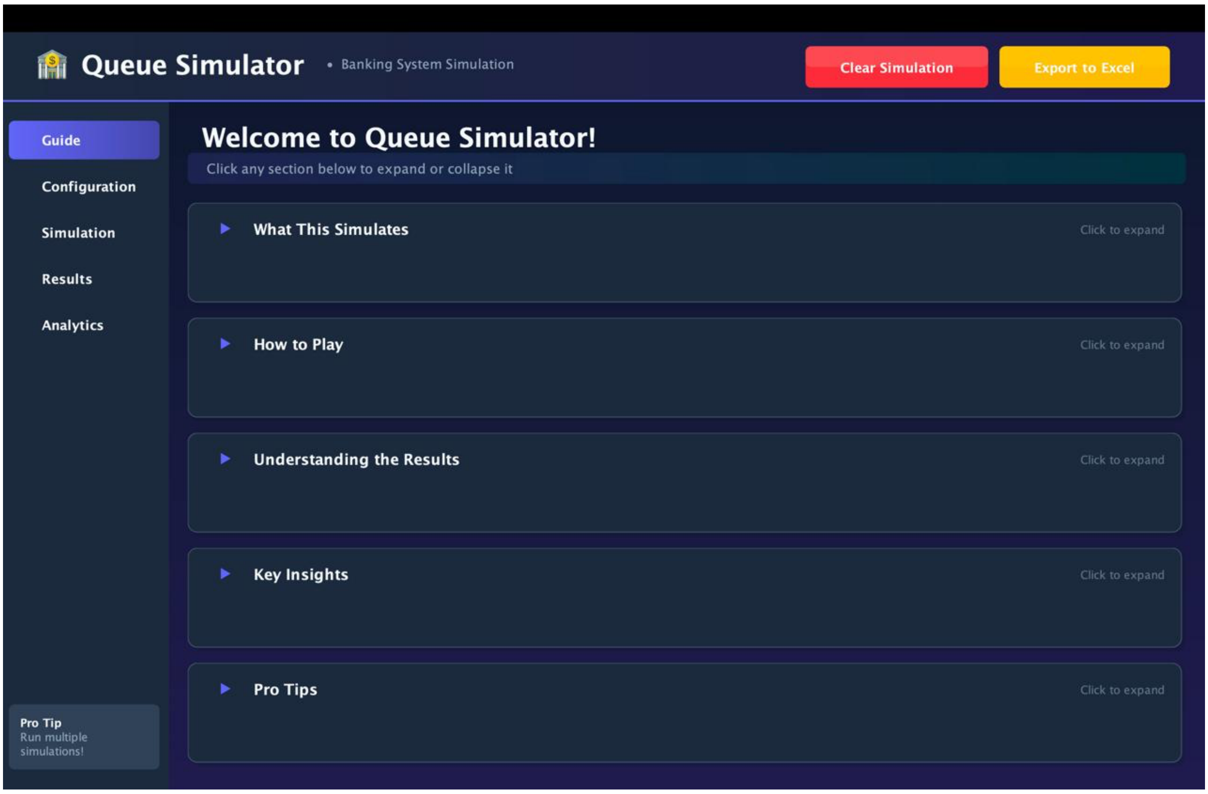

Figure 1 Guide page that explains how to use our simulation

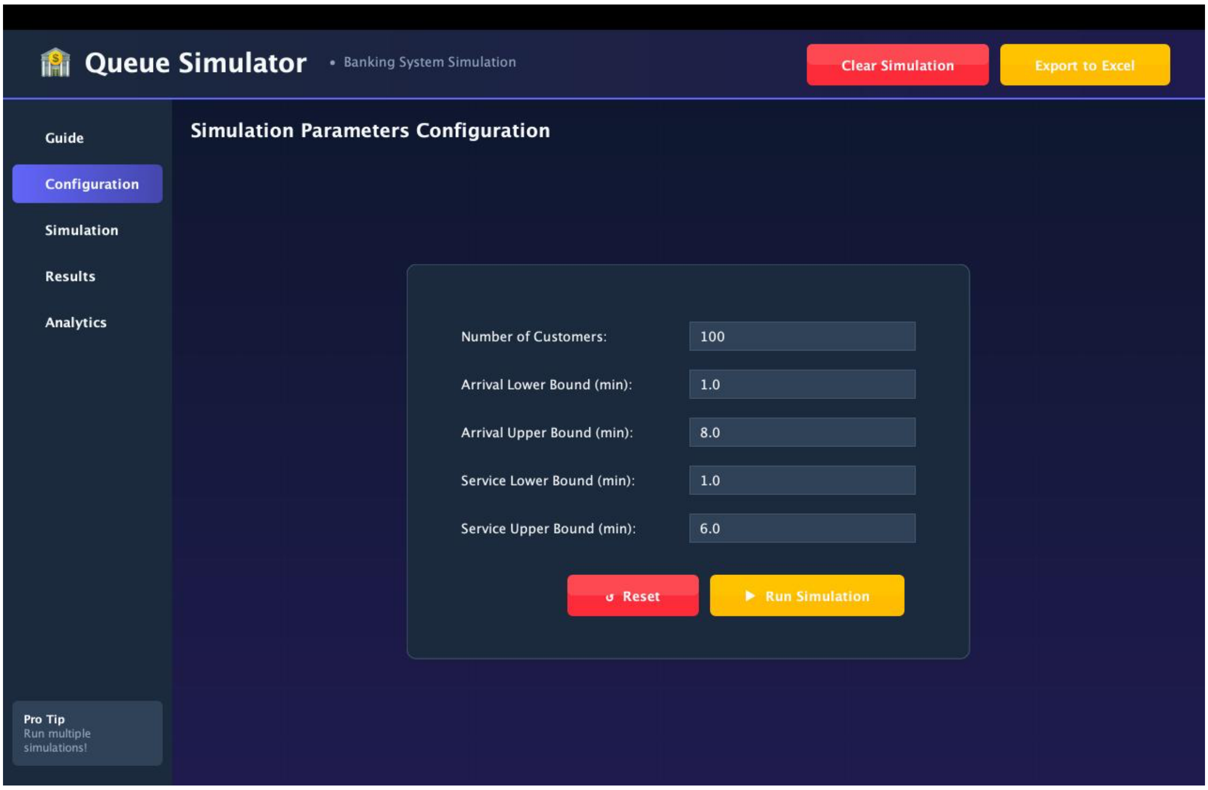

Figure 2 Configuration page to put custom settings

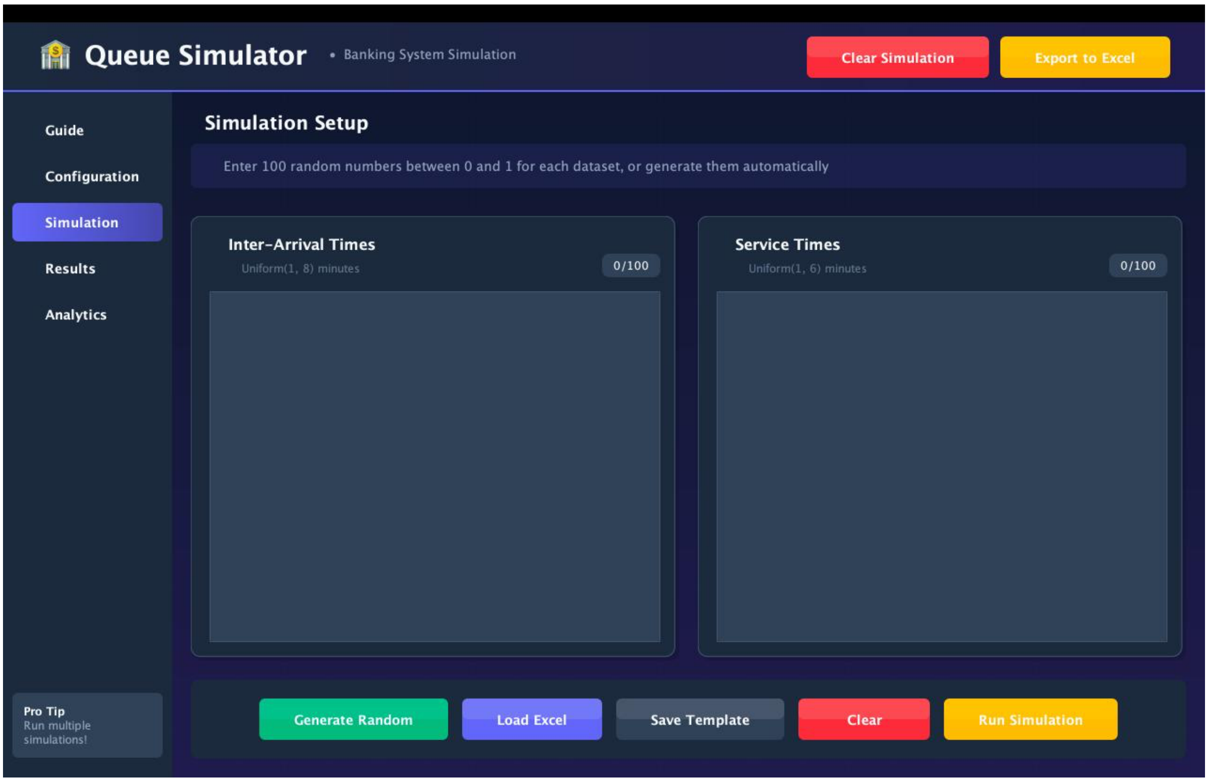

Figure 3 Simulation page that allows you to generate random value or import from a spreadsheet

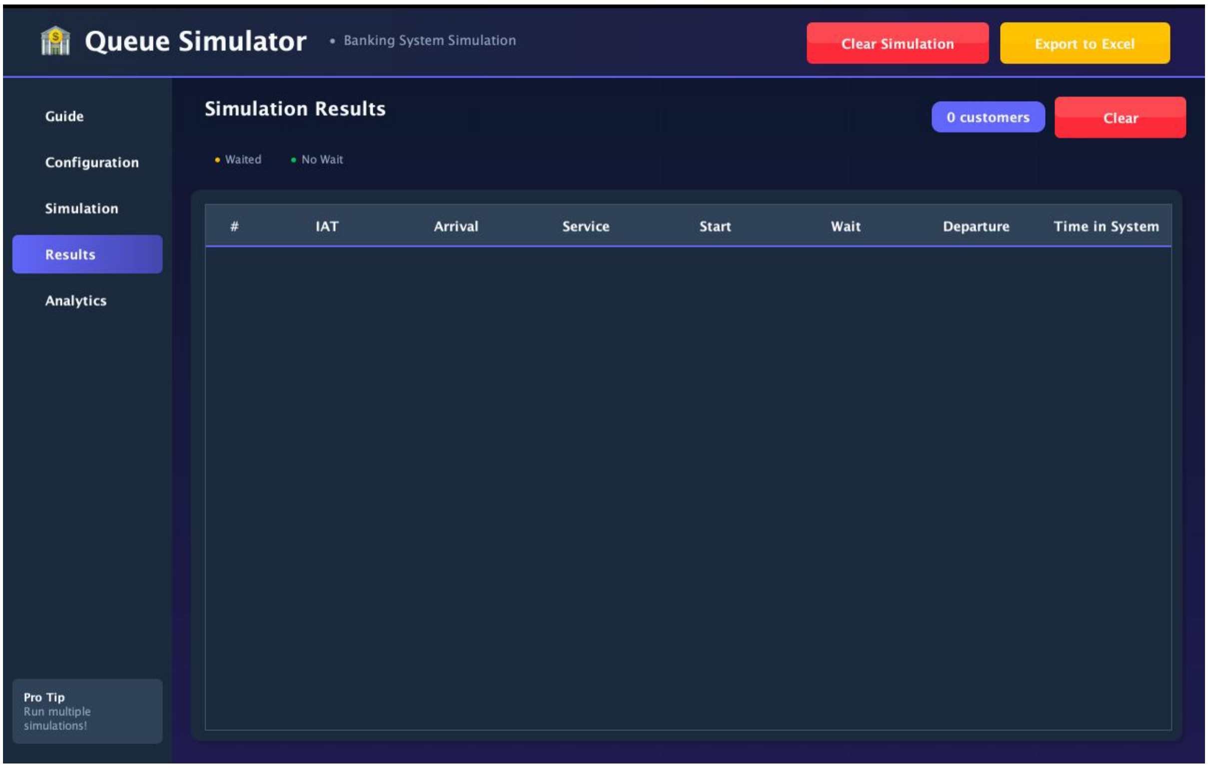

Figure 4 Results page where you can review the results of your simulation

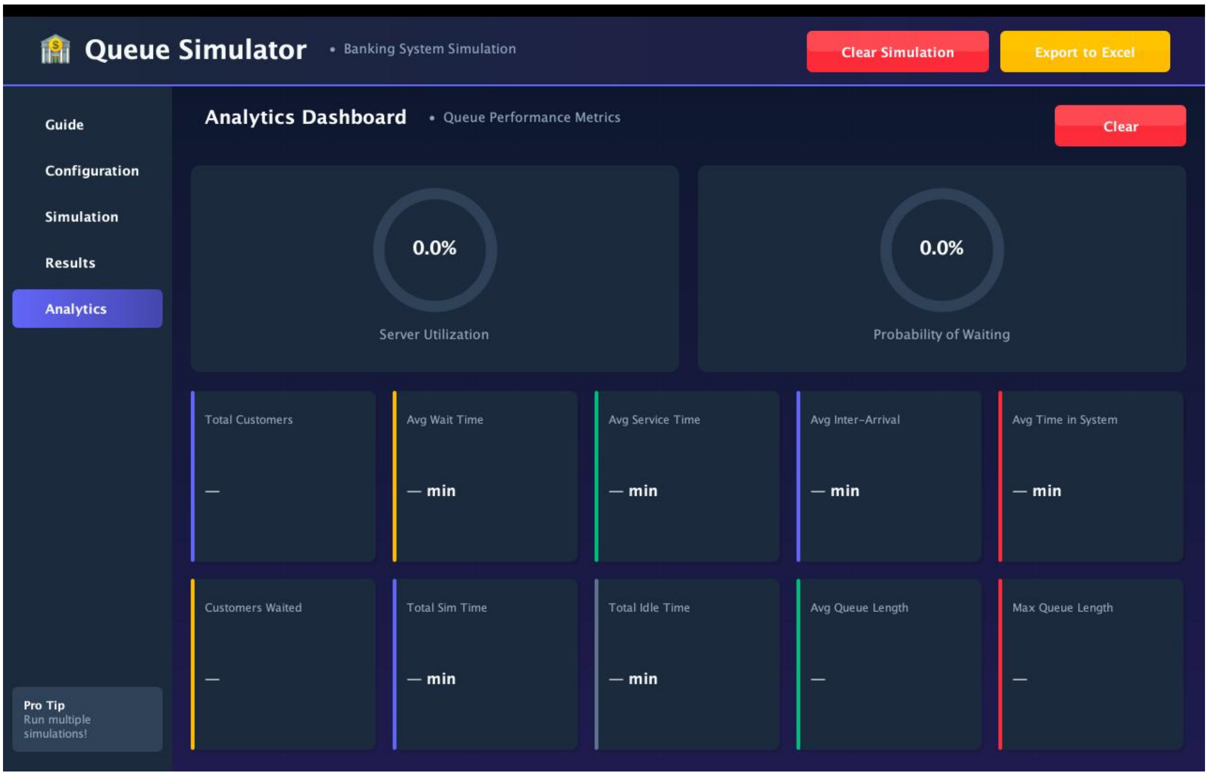

Figure 5 Analytics page where you can view different statistic of your simulation